COS80023 Big Data – Lab 10

Lab10: Credit Task 10 Clustering vs. Classifiers

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Classification output from RS classifier:

3 labels – 'setosa', 'versicolor', 'virginica'

```
> rf_pred_test
 [1] setosa     setosa     setosa     setosa     setosa     setosa     setosa     setosa
 [9] setosa     setosa     setosa     setosa     setosa     setosa     setosa     versicolor
[17] versicolor versicolor versicolor versicolor virginica  versicolor versicolor versicolor
[25] versicolor versicolor versicolor versicolor versicolor versicolor virginica  virginica
[33] versicolor virginica  virginica  virginica  virginica  virginica  virginica  virginica
[41] virginica  virginica  virginica  virginica  virginica
Levels: setosa versicolor virginica
```

Q1. What variable should be entered for each parameter, and why?

For k-means, the data input used only the four numeric measurements: sepal_length, sepal_width, petal_length, petal_width. The label species was left out because clustering was unsupervised and did not use labels. The number of clusters k was set to 3, matching the three known groups in the iris data (Setosa, Versicolor, Virginica) and allowing a direct comparison between clusters and species.

```
km  <- test  %>% select(sepal_length, sepal_width, petal_length, petal_width)

km_test  <- kmeans(km,3)
```

Q2. Should clustering use the train set, test set, or the complete dataset? Why?

Generally, clustering does not require labels, so can be used on any split. In this assignment, k-means was run on the test rows so that each row aligned with both its true species and the Random Forest (RF) prediction. This made a simple, row-by-row comparison possible between cluster assignment, model prediction, and the true label.

```
km  <- test  %>% select(sepal_length, sepal_width, petal_length, petal_width)
```

Q3. Did the k-means cluster assignments agree with the Random Forest predictions, and could clusters be identified as species?

```
> km_test$cluster
  6   9  10  16  21  26  30  38  39  42  44  45  48  49  50  52  58  60  62  69  73  75  80  85
  2   2   2   2   2   2   2   2   2   2   2   2   2   2   2   1   1   1   1   1   1   1   1   1
 86  88  89  91  92  96 101 102 107 109 112 113 119 122 124 131 139 140 141 148 150
  1   1   1   1   1   1   3   1   1   3   3   3   3   1   1   3   1   3   3   3   1
>
```

```
> rf_cm
Confusion Matrix and Statistics

            Reference
Prediction   setosa versicolor virginica
  setosa         15          0         0
  versicolor      0         14         1
  virginica       0          1        14

Overall Statistics

               Accuracy : 0.9556
                 95% CI : (0.8485, 0.9946)
    No Information Rate : 0.3333
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.9333

 Mcnemar's Test P-Value : NA

Statistics by Class:

                     Class: setosa Class: versicolor Class: virginica
Sensitivity                 1.0000            0.9333           0.9333
Specificity                 1.0000            0.9667           0.9667
Pos Pred Value              1.0000            0.9333           0.9333
Neg Pred Value              1.0000            0.9667           0.9667
Prevalence                  0.3333            0.3333           0.3333
Detection Rate              0.3333            0.3111           0.3111
Detection Prevalence        0.3333            0.3333           0.3333
Balanced Accuracy           1.0000            0.9500           0.9500
>
```

Agreement was **partial**, and clusters were identified by **majority vote**. The Random Forest (supervised) reached **95.56% accuracy** on the test set with **Kappa = 0.933**. Class results were **Sensitivity/Specificity** of **1.00/1.00 (Setosa)**, **0.933/0.967 (Versicolor)**, and **0.933/0.967 (Virginica)**.

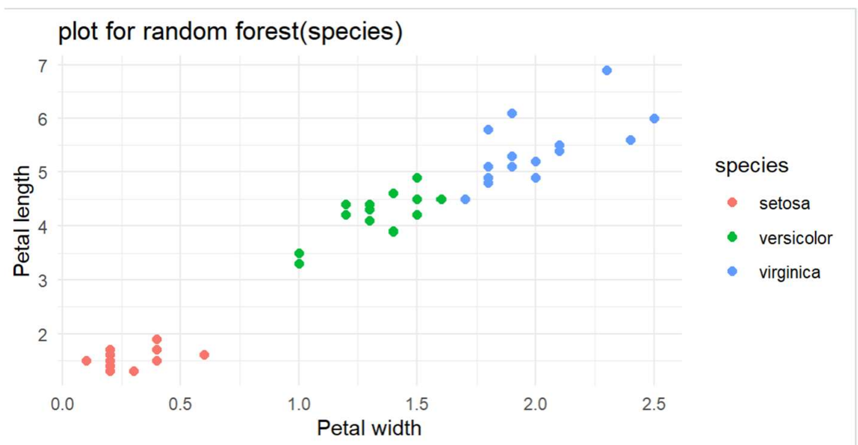
The k-means result (unsupervised), after mapping each cluster to the species most common within it, achieved **86.67% accuracy** on the same test rows. The observed mapping was **cluster 3 → Setosa**, **cluster 2 → Versicolor**, and **cluster 1 → Virginica**. Setosa was perfectly recovered (**15/15**). Most mistakes occurred between **Versicolor** and **Virginica** (e.g., **6 Virginica** points assigned to the Versicolor cluster). In summary, the supervised RF performed better overall, while k-means still recovered the main grouping structure, especially the clearly separated Setosa class.

```
> table_RF_vs_true <- table(cmp2$rf_pred, cmp2$species)
> table_KM_vs_true <- table(cmp2$mapped_species, cmp2$species)
> table_RF_vs_true

             setosa versicolor virginica
  setosa         15          0         0
  versicolor      0         14         1
  virginica       0          1        14
> table_KM_vs_true

             setosa versicolor virginica
  setosa         15          0         0
  versicolor      0         15         6
  virginica       0          0         9
>
> rf_acc <- mean(cmp2$rf_pred == cmp2$species)
> km_acc <- mean(cmp2$mapped_species == cmp2$species)
> rf_acc; km_acc
[1] 0.9555556
[1] 0.8666667
> |
```

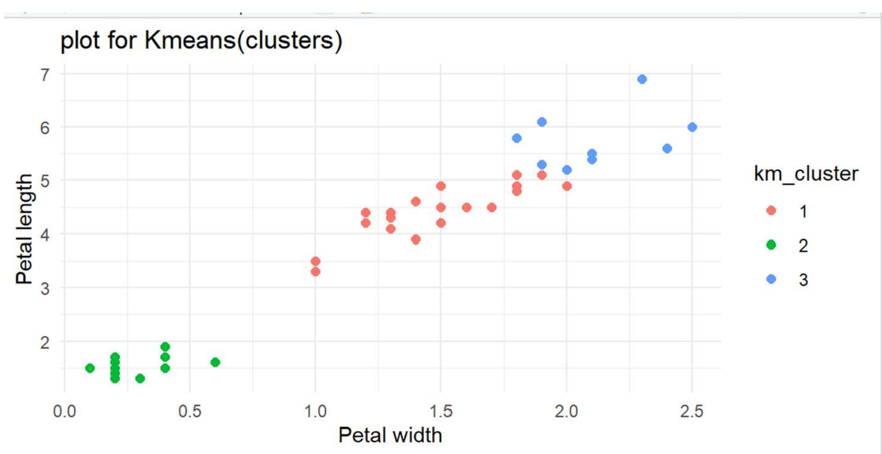
Q4. Why did kmeans() assign the clusters as observed?

Petal length vs Petal Size Scatter plots

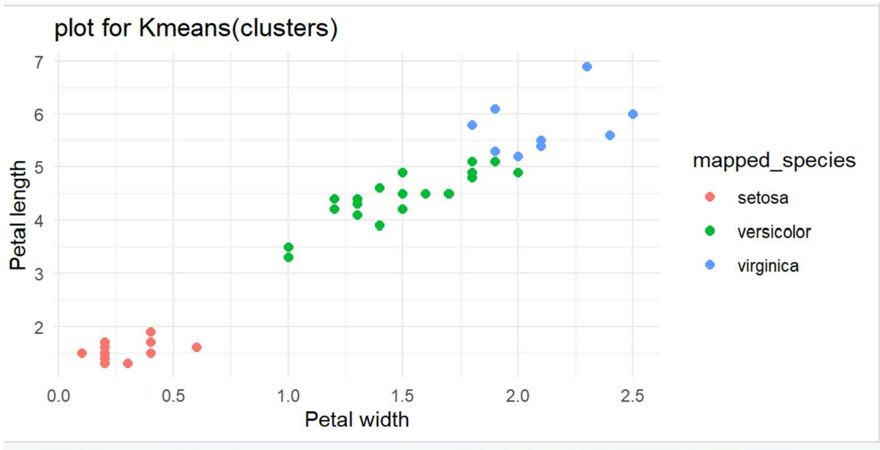
Petal length vs petal width was the best scatter plot to compare and pick the differences



RF classier scatter plot



kmeans clustering scatter plot with cluster id



kmeans clustering scatter plot with mapped species to respective cluster IDs

The result followed directly from the shapes and positions of the points in feature space.

One group had very small petal length and petal width, matching Setosa, and was clearly separate; k-means grouped these points with no confusion. The other two species, Versicolor and Virginica, occupied a shared region with larger petals and only subtle differences, so their values **overlapped**. Because k-means places **centers (centroids)** and assigns each point to the **nearest** one, it effectively draws **round-ish regions** around those centers. In the overlapped Versicolor/Virginica area, no single round boundary matched the true species split, so k-means divided that shared region into two clusters that were close, but not identical, to the species boundary. This produced the observed mix-ups (for example, six Virginica points grouped with Versicolor) and explains the performance gap: the supervised Random Forest, which learned the true labels, reached **95.56%** accuracy, while the label-free k-means reached **86.67%**. **Petal width and length were the main significant determining features for both clustering and classification.**

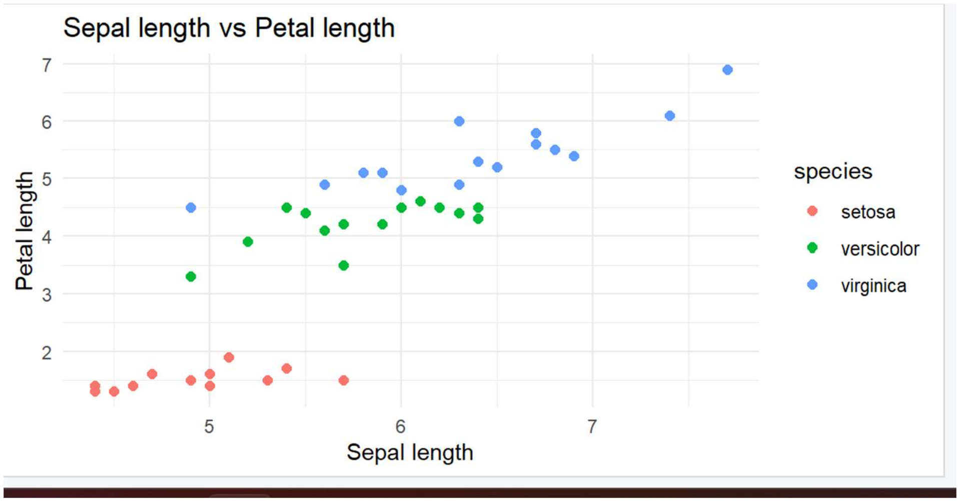
Also, checked out other independent variables combination scatterplots to check.

petal_length vs sepal_length and **petal_width vs sepal_width** are good alternates that still show meaningful class structure.

Sepal-only views are mainly illustrative; they do not offer much information for clustering or classification, even though the sepal width and length are taken into account by the models.

Sepal length vs Petal length scatterplots

For RF classification



For K means Clustering